Week-1

Program-1:

```
#include <stdio.h>
int main(){
printf("Welcome to operating systems\n");
return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ nano hello.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc hello.c -o hello
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./hello
Welcome to operating systems
```

Program-2:

```
tejaswini@LAPTOP-S8US186U

  GNU nano 8.7.1
#include<stdio.h>
int main(){
char name[30];
printf("Enter your name:");
scanf("%s",name);
printf("Welcome %s\n",name);
return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc readusername.c -o readusername
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./readusername
Enter your name:Tejaswini
Welcome Tejaswini
```

Program-3:

```
tejaswini@LAPTOP-S8US186U

  GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
int main(){
printf("Process ID = %d\n",getpid());
return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc displayprocessID.c -o displayprocessID
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./displayprocessID
Process ID = 1037
tejaswini@LAPTOP-S8US186U:~/OS_Lab$
```

Program-4:

```
  GNU nano 8.7.1
#include<stdio.h>
int main(){
fork();
printf("Hello\n");
return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc fork.c -o fork
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./fork
Hello
Hello
tejaswini@LAPTOP-S8US186U:~/OS_Lab$
```

Program-5:

```
  GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
int main(){
pid_t pid;
pid=fork();
if(pid==0)
{
printf("I am Child Process\n");
}
else
{
printf("I am Parent Process\n");
}
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ nano parentandchild.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc parentandchild.c -o parentandchild
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./parentandchild
I am Parent Process
I am Child Process
```

Program-6:

```
  GNU nano 8.7.1                                                              parentandchild2.c
#include<stdio.h>
#include<unistd.h>
int main(){
pid_t pid;
pid=fork();
if(pid==0){
printf("Child PID = %d\n",getpid());
printf("Parent PID = %d\n",getppid());
}
else{
printf("Parent PID = %d\n",getpid());
}}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc parentandchild2.c -o parentandchild2
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./parentandchild2
Parent PID = 700
Child PID = 701
Parent PID = 329
tejaswini@LAPTOP-S8US186U:~/OS_Lab$
```

Program-7:

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>
int main(){
pid_t pid;
pid=fork();
if(pid==0){
printf("Child Running\n");
}
else{
wait(NULL);
printf("Parent Running After Child\n");
}
return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc wait.c -o wait
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./wait
Child Running
Parent Running After Child
tejaswini@LAPTOP-S8US186U:~/OS_Lab$
```